Quantum Mechanics

Lecture 8: Hamiltonian Time Evolution, Interference, and the Schrödinger Equation

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Chapter 1

Hamiltonian Time Evolution, Interference, and the Schrödinger Equation

1.1 The Classical Hamiltonian Generates a Flow

We now begin the serious study of change with time. It is useful to start with the classical picture, not because we shall need every detail of classical mechanics, but because one structural idea survives intact: the Hamiltonian is the generator of time evolution.

For a system with finitely many classical states, a law of evolution may be drawn as arrows from one state to the next. A coin could pass through a sequence such as heads, tails, heads, and so on. Since there is no continuous path between two isolated states, this picture naturally describes updates at discrete sampling times.

Continuous classical motion requires a continuous state space. A particle on a line has, at each instant, a position x and momentum p. Its state is therefore a point in the two-dimensional phase space with coordinates (x, p). As time passes, that point follows a flow line.

The flow is determined by a function of phase space called the Hamiltonian. For a particle of mass m in a potential $V(x)$, the standard example is

$$H(x,p) = \frac{p^2}{2m} + V(x). \tag{1.1}$$

The momentum-dependent term is the kinetic energy and the position-dependent term is the potential energy. For many particles there is one momentum conjugate to each coordinate, but one pair is enough to display the structure.

Hamilton's equations turn this energy function into motion:

$$\dot{x} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial x}. \tag{1.2}$$

For Equation (1.1), $\partial H/\partial p = p/m$, which is the velocity. The second equation states that the rate of change of momentum is the force, $-\partial V/\partial x$.

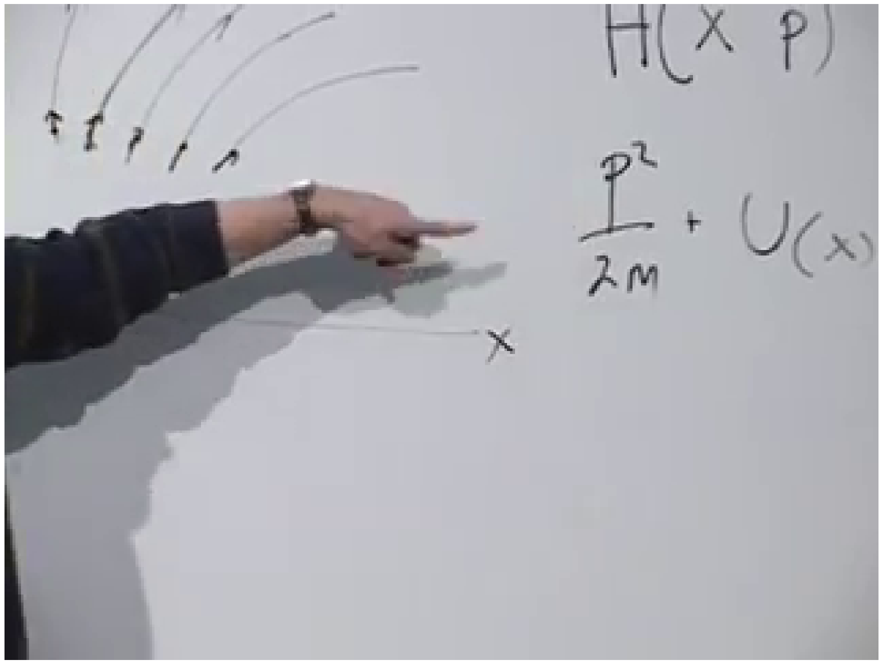

Figure 1.1: The classical Hamiltonian for a particle on a line, with the kinetic and potential terms separated.

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Now let $F(x, p)$ be any classical observable. The chain rule gives

$$\frac{dF}{dt} = \frac{\partial F}{\partial x}\dot{x} + \frac{\partial F}{\partial p}\dot{p} \tag{1.3}$$

$$= \frac{\partial F}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial H}{\partial x}. \tag{1.4}$$

The combination on the right is the Poisson bracket:

$$\{F, H\}_{\mathrm{PB}} = \frac{\partial F}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial F}{\partial p}\frac{\partial H}{\partial x}. \tag{1.5}$$

Thus

$$\boxed{\frac{dF}{dt} = \{F, H\}_{\mathrm{PB}}.} \tag{1.6}$$

One need not memorize the bracket at this stage. Its message is more important than its detailed form: once H is known, every phase-space quantity has a determined time derivative. The Hamiltonian generates the flow.

1.2 Quantum Evolution Preserves Relations Between States

The quantum statement begins from a different picture. A state is a vector, and the physically meaningful relation between two normalized states is their inner product. Orthogonal states can be distinguished with certainty; nonorthogonal states have a nonzero transition probability. Closed-system evolution must preserve all of these relations.

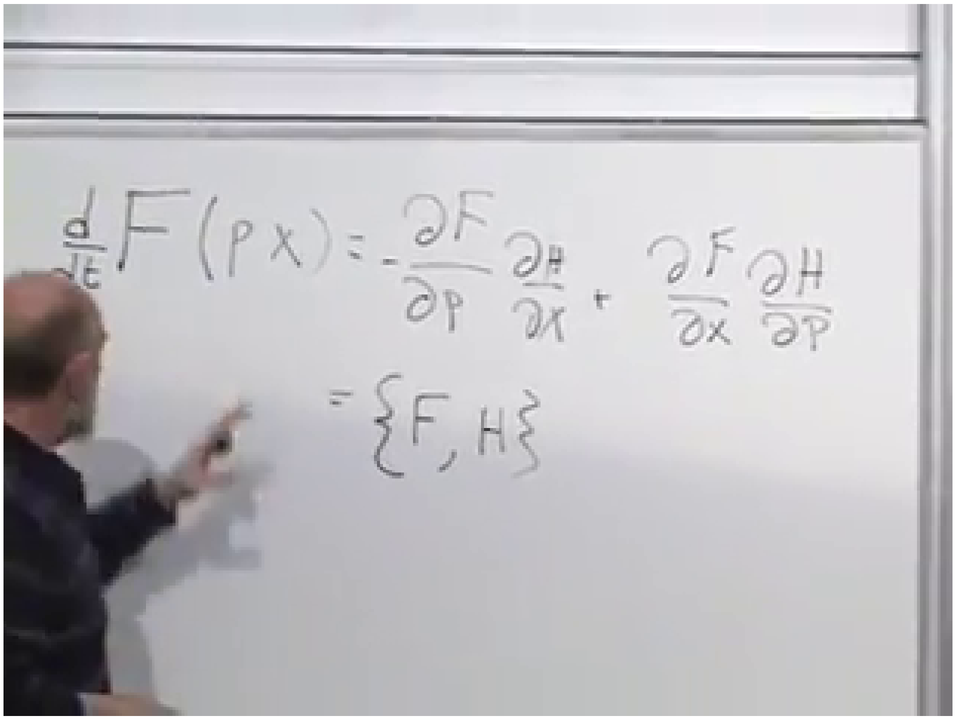

Figure 1.2: Hamilton's equations inserted into the chain rule produce the Poisson-bracket law for an arbitrary observable.

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Suppose that after an elapsed time T,

$$|A\rangle \longmapsto \hat{U}(T)\,|A\rangle\,, \qquad |B\rangle \longmapsto \hat{U}(T)\,|B\rangle\,. \tag{1.7}$$

The corresponding transformed bra is

$$\langle B| \longmapsto \langle B|\,\hat{U}^\dagger(T). \tag{1.8}$$

Preservation of the inner product requires

$$\langle B \mid A\rangle = \langle B|\,\hat{U}^\dagger(T)\hat{U}(T)\,|A\rangle\,. \tag{1.9}$$

This must hold for every pair $|A\rangle\,, |B\rangle$. Therefore

$$\boxed{\hat{U}^\dagger(T)\hat{U}(T) = \mathbf{1}.} \tag{1.10}$$

In finite dimensions this also implies $\hat{U}(T)\hat{U}^\dagger(T) = \mathbf{1}$. The evolution operator is unitary.

There is a useful geometric image here. The vectors move, but the whole state space moves rigidly: norms and mutual angles remain fixed. In particular, orthogonal states remain orthogonal. For the linear evolution considered here, this preservation law is the governing principle.

Writing the state at an arbitrary initial time t, the evolution law is

$$|\psi(t+T)\rangle = \hat{U}(T)\,|\psi(t)\rangle\,. \tag{1.11}$$

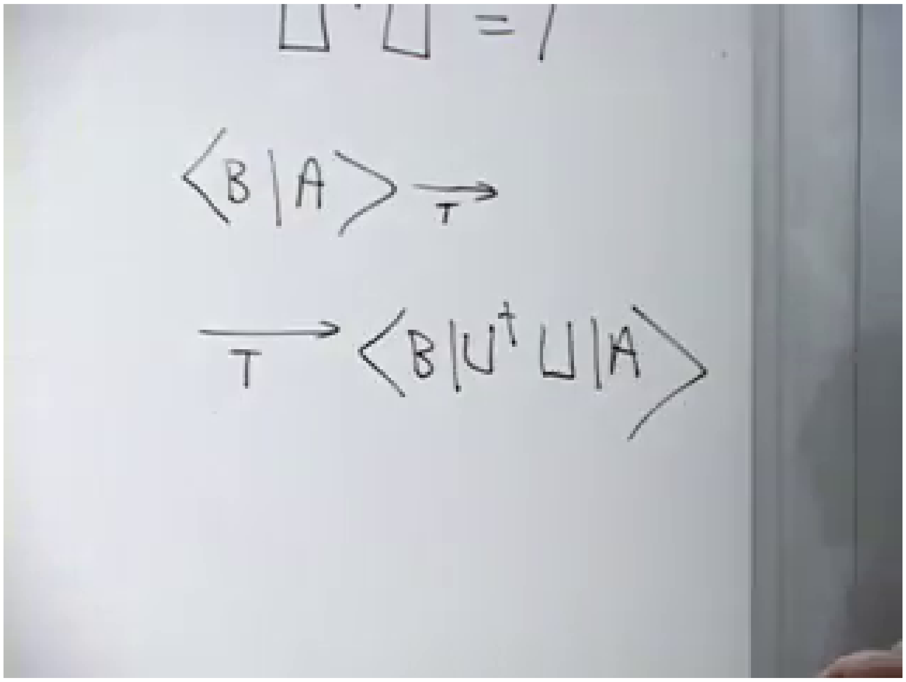

Figure 1.3: The common evolution of two states preserves their inner product and leads to $\hat{U}^\dagger \hat{U} = \mathbf{1}$.
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No elapsed time means no change,

$$\hat{U}(0) = \mathbf{1}. \tag{1.12}$$

Successive intervals compose:

$$\hat{U}(T_2)\hat{U}(T_1) = \hat{U}(T_1 + T_2) \tag{1.13}$$

when the Hamiltonian has no explicit time dependence. This is why a long evolution can be assembled from many short steps.

1.3 The Infinitesimal Generator

Let ϵ be short enough that terms of order ϵ^2 may be neglected. Since $\hat{U}(0) = \mathbf{1}$, the most general first-order form can be written

$$\hat{U}(\epsilon) = \mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H} + O(\epsilon^2). \tag{1.14}$$

At first, $\hat{H}$ is simply the name of an operator. The factors $-i/\hbar$ are chosen so that $\hat{H}$ has units of energy and so that the classical correspondence will later take its standard form.

Taking the adjoint gives

$$\hat{U}^\dagger(\epsilon) = \mathbf{1} + \frac{i\epsilon}{\hbar}\hat{H}^\dagger + O(\epsilon^2). \tag{1.15}$$

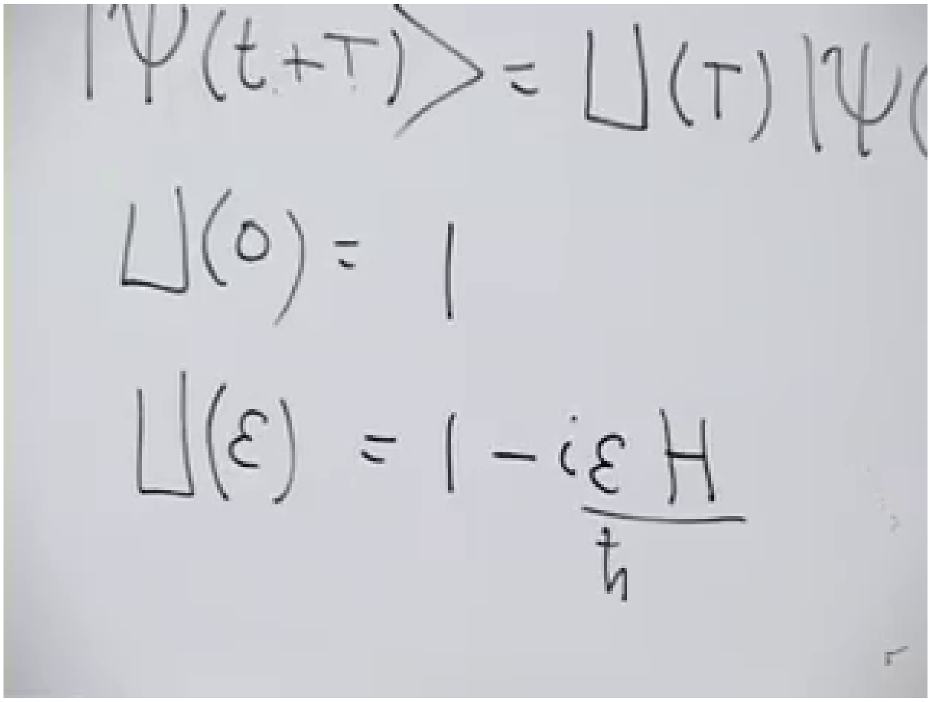

Figure 1.4: A short-time evolution differs from the identity by a term proportional to $-i\epsilon\hat{H}/\hbar$.
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Unitarity through first order then says

$$\begin{aligned}
\mathbf{1} &= \hat{U}^\dagger(\epsilon)\hat{U}(\epsilon) && (1.16)\\
&= \left(\mathbf{1} + \frac{i\epsilon}{\hbar}\hat{H}^\dagger\right)\left(\mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H}\right) + O(\epsilon^2) && (1.17)\\
&= \mathbf{1} + \frac{i\epsilon}{\hbar}\left(\hat{H}^\dagger - \hat{H}\right) + O(\epsilon^2). && (1.18)
\end{aligned}$$

The linear term must vanish, so

$$\boxed{\hat{H}^\dagger = \hat{H}.} \tag{1.19}$$

The generator of continuous unitary evolution is Hermitian. Its eigenvalues are real and, in finite dimensions, its eigenvectors may be chosen as an orthonormal basis. We identify this observable as the Hamiltonian and call its eigenvalues energies. At this point that is an identification within quantum mechanics; recovering familiar classical energy requires a further correspondence argument.

For a time-independent Hamiltonian, compounding the short-time step gives

$$\hat{U}(T) = \exp\left(-\frac{i}{\hbar}\hat{H}T\right). \tag{1.20}$$

The exponential is unitary precisely because $\hat{H}$ is Hermitian.

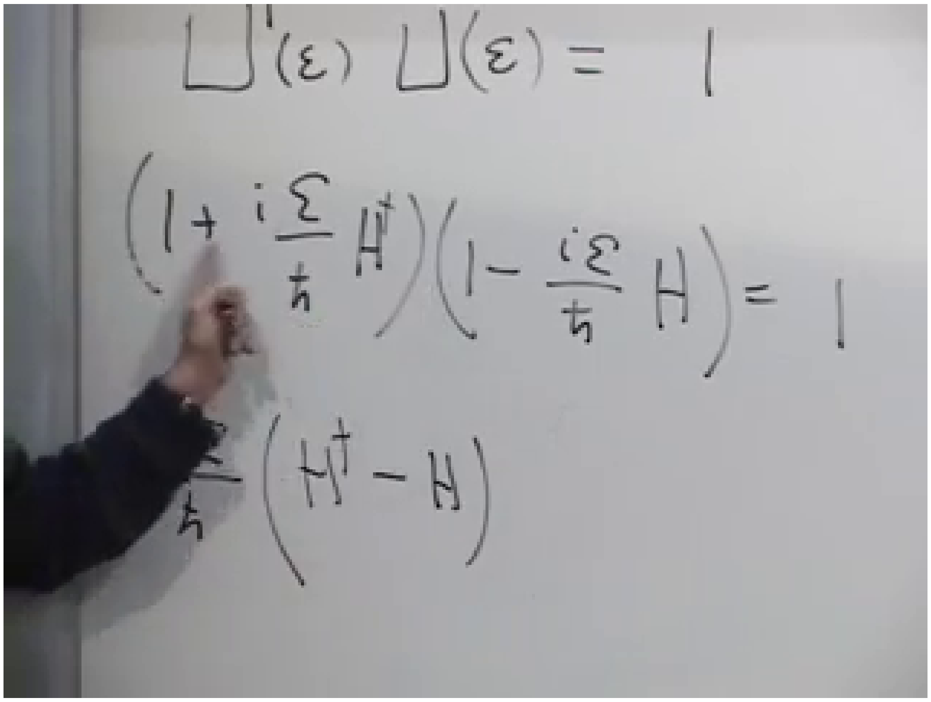

Figure 1.5: Expanding $\hat{U}^\dagger \hat{U} = \mathbf{1}$ to first order leaves the condition $\hat{H}^\dagger - \hat{H} = 0$.
Lecture frame: 00:22:48.

Question from the audience. If $\hbar$ appears in the denominator, what happens in the classical limit $\hbar \to 0$? [a]

Response. The classical limit is not obtained by setting a denominator to zero inside one fixed microscopic formula. What matters are dimensionless actions measured in units of $\hbar$, such as $\epsilon E/\hbar$. Classical behavior emerges when the relevant actions are large compared with $\hbar$ and quantum phases combine by stationary phase or decoherence. The operator $\hat{H}$ itself has a well-defined classical counterpart even though the phases become rapidly varying.

[a] Lecture timestamp: 00:26:41–00:28:07.

1.4 The Abstract Schrödinger Equation

Apply Equation (1.14) to a state:

$$|\psi(t+\epsilon)\rangle = \left(\mathbf{1} - \frac{i\epsilon}{\hbar}\hat{H}\right)|\psi(t)\rangle + O(\epsilon^2). \tag{1.21}$$

Subtract $|\psi(t)\rangle$, divide by ϵ, and take $\epsilon \to 0$:

$$\frac{\partial}{\partial t}|\psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\psi(t)\rangle. \tag{1.22}$$

Multiplication by $i\hbar$ gives

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle.} \tag{1.23}$$

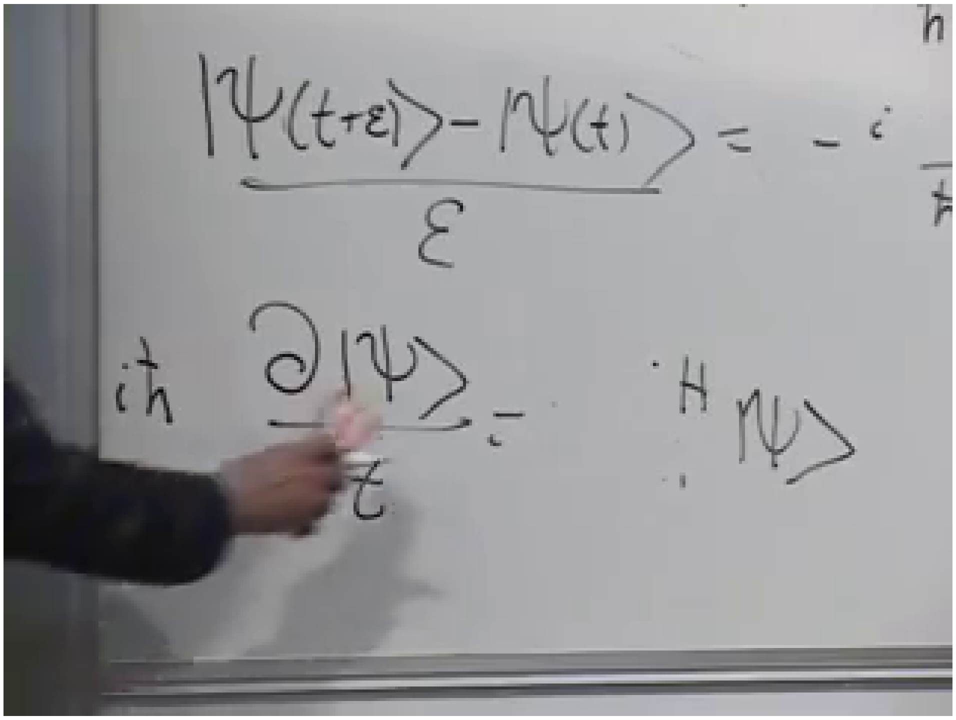

Figure 1.6: The finite difference from t to $t+\epsilon$ becomes the abstract Schrödinger equation.
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This is the general law of quantum time evolution. It is abstract because $|\psi\rangle$ may describe polarization, spin, a particle in space, or any other quantum system. Schrödinger's original wave equation will appear only after we choose a coordinate representation and specify $\hat{H}$.

There is a useful parallel with momentum. In the coordinate representation,

$$\hat{p} = -i\hbar \frac{\partial}{\partial x} \tag{1.24}$$

relates spatial translation to momentum, while Equation (1.23) relates time translation to energy. The sign conventions are conventional; their consistent use is what matters.

1.5 A Polarization Doublet in an Anisotropic Medium

1.5.1 The Energy Eigenstates

Consider a photon of fixed wavelength moving through a material whose two preferred polarization axes have different energies. Choose those material axes as x and y, and suppose

$$\hat{H}\,|x\rangle = E_1\,|x\rangle\,, \qquad \hat{H}\,|y\rangle = E_2\,|y\rangle\,. \tag{1.25}$$

Here x and y label polarization, not position. The assumption is that the material itself distinguishes these two directions.

A general polarization state is

$$|\psi(t)\rangle = \alpha(t)\,|x\rangle + \beta(t)\,|y\rangle = \begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix}. \tag{1.26}$$

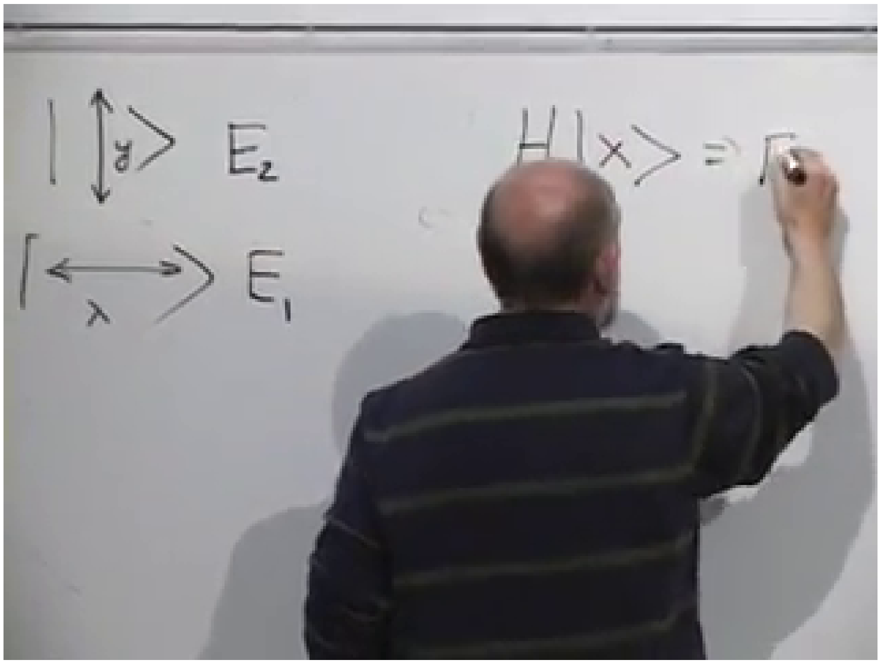

Figure 1.7: The material assigns energies E_1 and E_2 to its two orthogonal polarization eigenstates.
Lecture frame: 00:35:54.

The probabilities of obtaining x and y polarization are $|\alpha(t)|^2$ and $|\beta(t)|^2$.

In the $(|x\rangle, |y\rangle)$ basis,

$$\hat{H} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}. \tag{1.27}$$

The diagonal order follows directly from Equation (1.25).

The Schrödinger equation becomes

$$i\hbar \frac{d}{dt} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{1.28}$$

The two component equations decouple:

$$i\hbar\dot{\alpha} = E_1\alpha, \qquad i\hbar\dot{\beta} = E_2\beta. \tag{1.29}$$

Their solutions are

$$\alpha(t) = \alpha(0)e^{-iE_1 t/\hbar}, \qquad \beta(t) = \beta(0)e^{-iE_2 t/\hbar}. \tag{1.30}$$

Thus

$$|\psi(t)\rangle = \begin{pmatrix} \alpha(0)e^{-iE_1 t/\hbar} \\ \beta(0)e^{-iE_2 t/\hbar} \end{pmatrix}. \tag{1.31}$$

If we measure in the energy basis itself, nothing seems to change:

$$|\alpha(t)|^2 = |\alpha(0)|^2, \qquad |\beta(t)|^2 = |\beta(0)|^2. \tag{1.32}$$

Each exponential has unit modulus. The relative phase is invisible to an x/y analyzer, but it is not physically irrelevant.

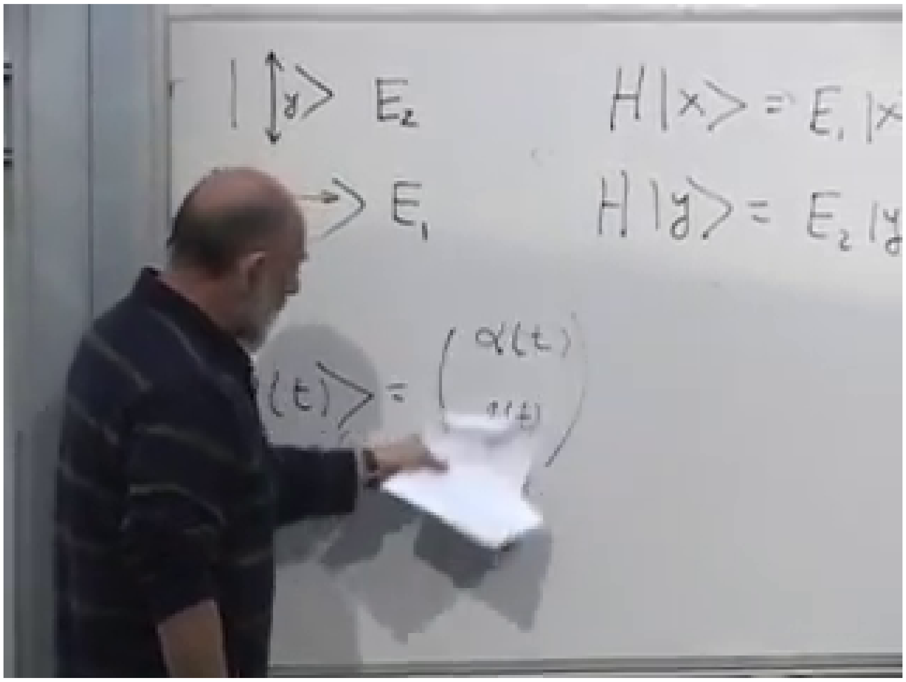

Figure 1.8: The polarization eigenvalue equations used to construct the diagonal Hamiltonian.

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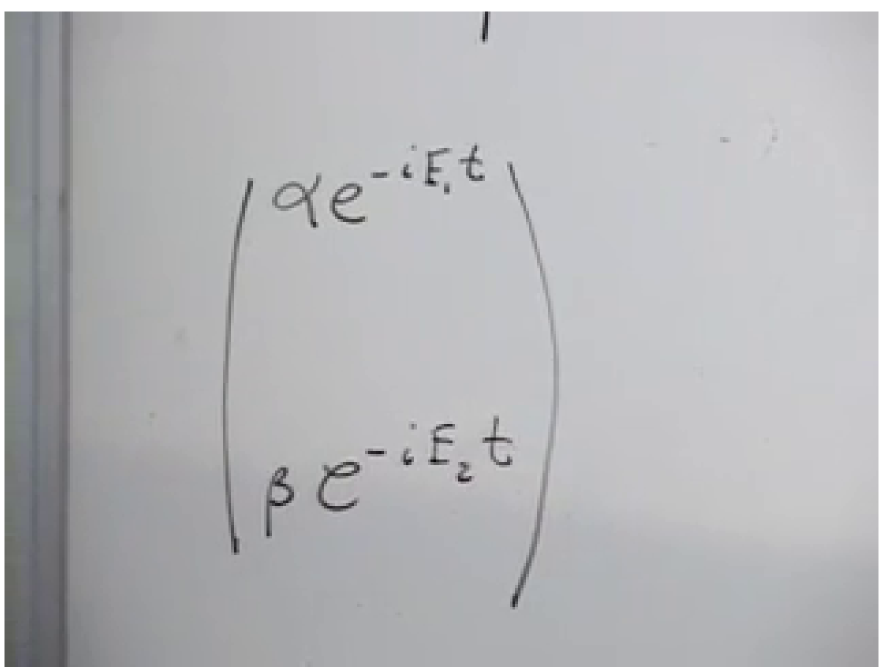

Figure 1.9: Each energy component accumulates its own phase, $e^{-iE_jt/\hbar}$.

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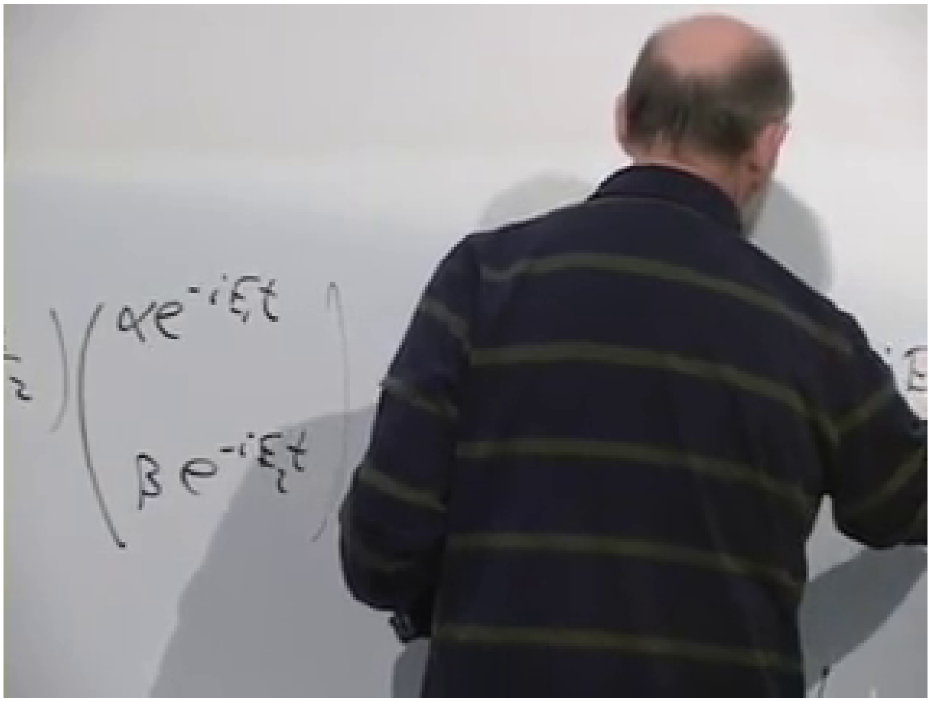

Figure 1.10: Projection of the evolved two-component state onto the 45° analyzer.

Lecture frame: 00:49:24.

1.5.2 Interference in a Rotated Analyzer

Let

$$|d_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |d_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \tag{1.33}$$

be the $+45°$ and $-45°$ linear polarizations. The amplitude to pass a $+45°$ analyzer after time t is

$$\langle d_+ \,|\, \psi(t)\rangle = \frac{1}{\sqrt{2}}\left[\alpha(0)e^{-iE_1t/\hbar} + \beta(0)e^{-iE_2t/\hbar}\right]. \tag{1.34}$$

Prepare the photon with a first $+45°$ polarizer. Then

$$\alpha(0) = \beta(0) = \frac{1}{\sqrt{2}}, \tag{1.35}$$

and

$$\mathcal{A}_+(t) = \frac{1}{2}\left(e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\right). \tag{1.36}$$

The transmission probability is the absolute square:

$$\begin{aligned}
P_+(t) &= |\mathcal{A}_+(t)|^2 && (1.37)\\
&= \frac{1}{4}\left(e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\right)\left(e^{iE_1t/\hbar} + e^{iE_2t/\hbar}\right) && (1.38)\\
&= \frac{1}{4}\left[2 + e^{i(E_2-E_1)t/\hbar} + e^{-i(E_2-E_1)t/\hbar}\right]. && (1.39)
\end{aligned}$$

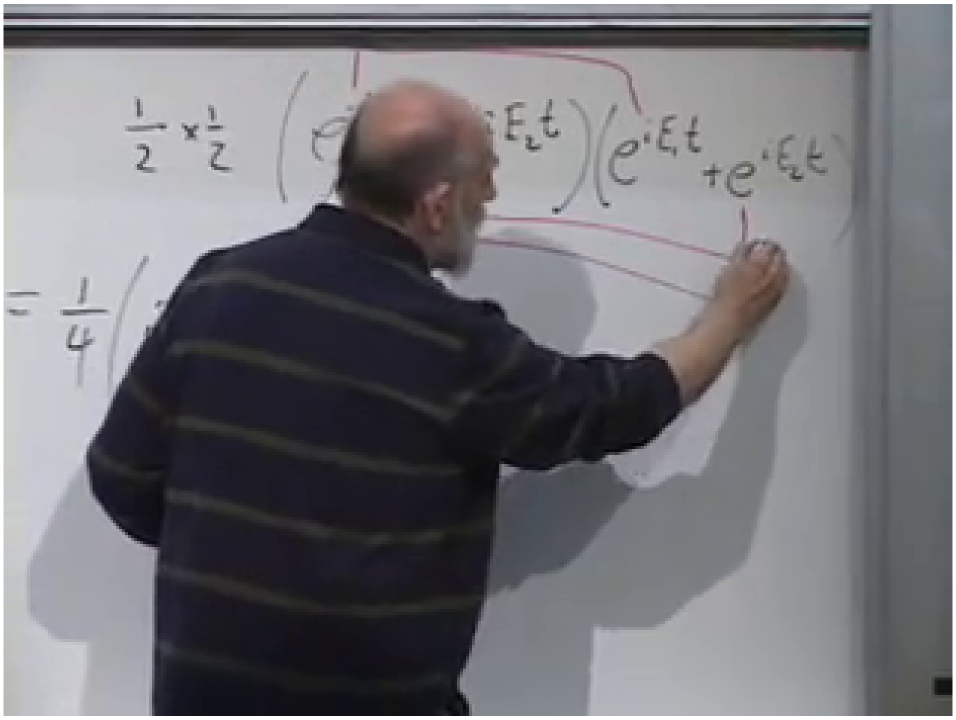

Figure 1.11: The absolute square contains two direct terms and two phase-sensitive cross terms.
Lecture frame: 00:54:54.

Define the energy splitting

$$\Delta E = E_2 - E_1. \tag{1.40}$$

Using $e^{i\delta} + e^{-i\delta} = 2\cos\delta$, we obtain

$$P_+(t) = \frac{1}{2}\left[1 + \cos\left(\frac{\Delta E\, t}{\hbar}\right)\right] \tag{1.41}$$

$$= \cos^2\left(\frac{\Delta E\, t}{2\hbar}\right). \tag{1.42}$$

The complementary analyzer has

$$P_-(t) = \sin^2\left(\frac{\Delta E\, t}{2\hbar}\right), \qquad P_+(t) + P_-(t) = 1. \tag{1.43}$$

At $t = 0$, two adjacent parallel analyzers transmit with certainty. When $\Delta E\, t/\hbar = \pi$, the state is orthogonal to its initial $+45^\circ$ polarization and passes the -45° analyzer with certainty.

Between those two times the state is generally not linearly polarized. Removing a common phase from Equation (1.31) gives

$$|\psi(t)\rangle \sim \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ e^{-i\Delta E t/\hbar} \end{pmatrix}. \tag{1.44}$$

It begins at $+45^\circ$ linear polarization, becomes circular when the relative phase is $\pm\pi/2$, reaches -45° linear polarization at π, and passes through elliptical states in between. The analyzer oscillation is therefore an interference measurement of a continuously changing polarization state.

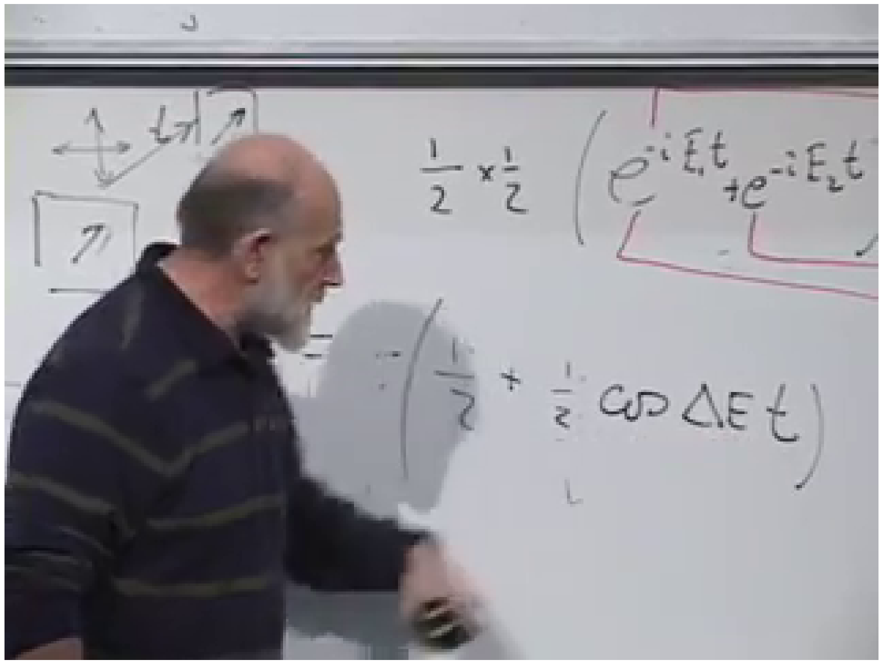

Figure 1.12: The probability for the second parallel analyzer oscillates between one and zero as the relative phase advances.

Lecture frame: 00:58:44.

1.5.3 Only Energy Differences Are Observable

If a constant C is added to every energy,

$$\hat{H} \longmapsto \hat{H} + C\mathbf{1}, \tag{1.45}$$

then

$$\hat{U}(t) \longmapsto e^{-iCt/\hbar}\hat{U}(t). \tag{1.46}$$

Every state acquires the same overall phase, which changes no transition probability. This is why the interference depends on ΔE, not on the arbitrary zero of energy.

Question from the audience. Must the material fill the entire region between the two polarizers?[a]

Response. The idealized calculation assumes that the photon spends a known time in a uniform anisotropic region. A finite slab works just as well: replace t by the transit time through that slab. Outside it, any common vacuum phase does not alter the polarization probabilities.

[a] Lecture timestamp: 01:00:52–01:01:27.

Question from the audience. The graph uses time on its horizontal axis. How does that become a downstream position in the experiment? [a]

Response. If the photon travels through the material with group velocity v, a slab length L corresponds to an interaction time $t = L/v$. Moving the second analyzer downstream, or changing the material thickness, therefore changes the accumulated relative phase $\Delta EL/(\hbar v)$.

[a] Lecture timestamp: 01:01:29–01:02:43.

Question from the audience. What happens if the incident photon is circularly polarized instead of linearly polarized?[a]

Response. A circular state is an equal-amplitude x/y superposition with relative phase $\pm\pi/2$. The material adds $-\Delta Et/\hbar$ to that phase, so the state moves through elliptical polarizations, becomes linearly polarized at appropriate times, and later reaches the opposite circular handedness. The same interference calculation applies after changing the initial coefficients.

[a]Lecture timestamp: 01:02:44–01:03:40.

Question from the audience. If the labels x and y are arbitrary, why should a different choice change the prediction? [a]

Response. The labels are conventional, but the axes are not. The material has two physical principal axes. We chose coordinates aligned with them so that $\hat{H}$ is diagonal. Rotating only the coordinate labels changes no prediction; rotating the polarizers relative to the material changes the preparation and measurement basis and therefore reveals the relative phase.

[a]Lecture timestamp: 01:03:43–01:05:15.

1.6 Expectation Values and Commutators

Let $\hat{K}$ be a fixed observable with no explicit time dependence. Its expectation value in the evolving state is

$$\left\langle \hat{K} \right\rangle = \langle \psi(t)| \hat{K} |\psi(t)\rangle . \tag{1.47}$$

Only the bra and ket vary. The product rule gives

$$\frac{d}{dt} \left\langle \hat{K} \right\rangle = \left\langle \dot{\psi} \right| \hat{K} |\psi\rangle + \langle \psi| \hat{K} \left| \dot{\psi} \right\rangle . \tag{1.48}$$

The ket equation and its adjoint are

$$\left| \dot{\psi} \right\rangle = -\frac{i}{\hbar} \hat{H} |\psi\rangle , \qquad \left\langle \dot{\psi} \right| = \frac{i}{\hbar} \langle \psi| \hat{H}, \tag{1.49}$$

where Hermiticity of $\hat{H}$ has been used.

Substitution gives

$$\frac{d}{dt} \left\langle \hat{K} \right\rangle = \frac{i}{\hbar} \langle \psi| \hat{H}\hat{K} |\psi\rangle - \frac{i}{\hbar} \langle \psi| \hat{K}\hat{H} |\psi\rangle \tag{1.50}$$

$$= \frac{i}{\hbar} \left\langle [\hat{H}, \hat{K}] \right\rangle \tag{1.51}$$

$$= -\frac{i}{\hbar} \left\langle [\hat{K}, \hat{H}] \right\rangle . \tag{1.52}$$

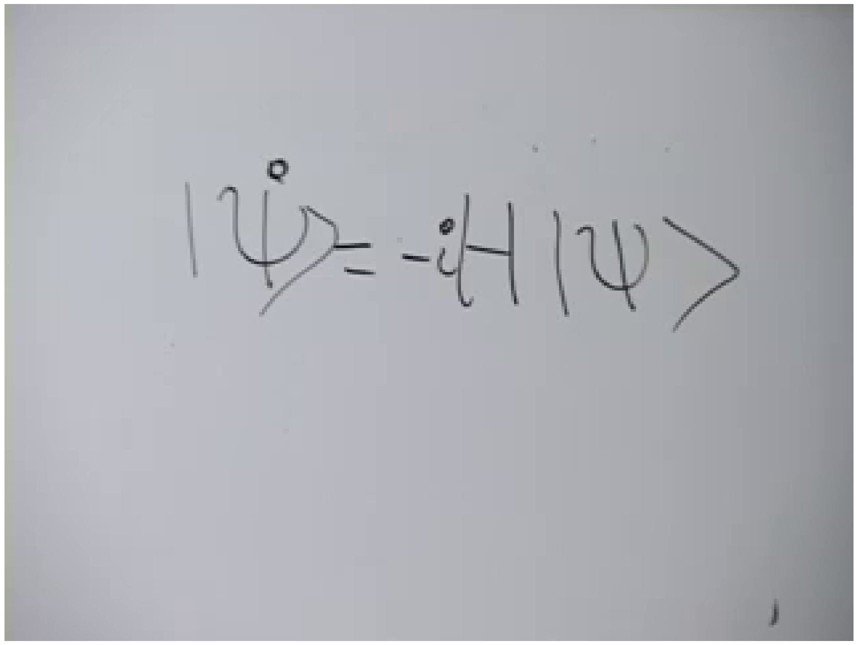

Figure 1.13: The ket Schrödinger equation and the corresponding bra equation carry opposite signs.

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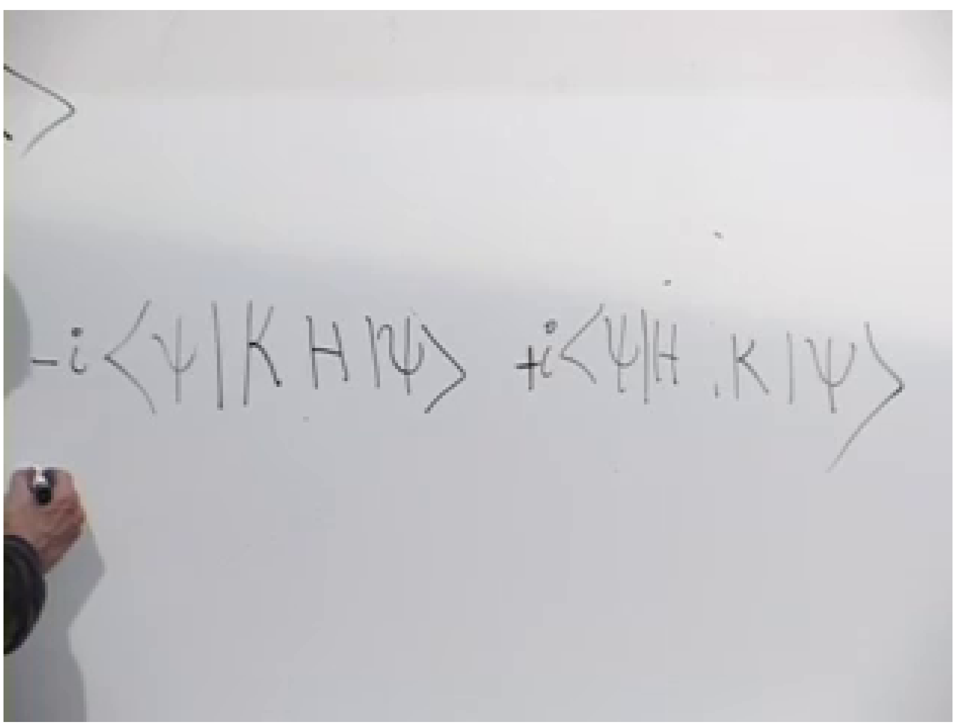

Figure 1.14: Differentiating the expectation-value sandwich produces the two operator orderings $\hat{H}\hat{K}$ and $\hat{K}\hat{H}$.

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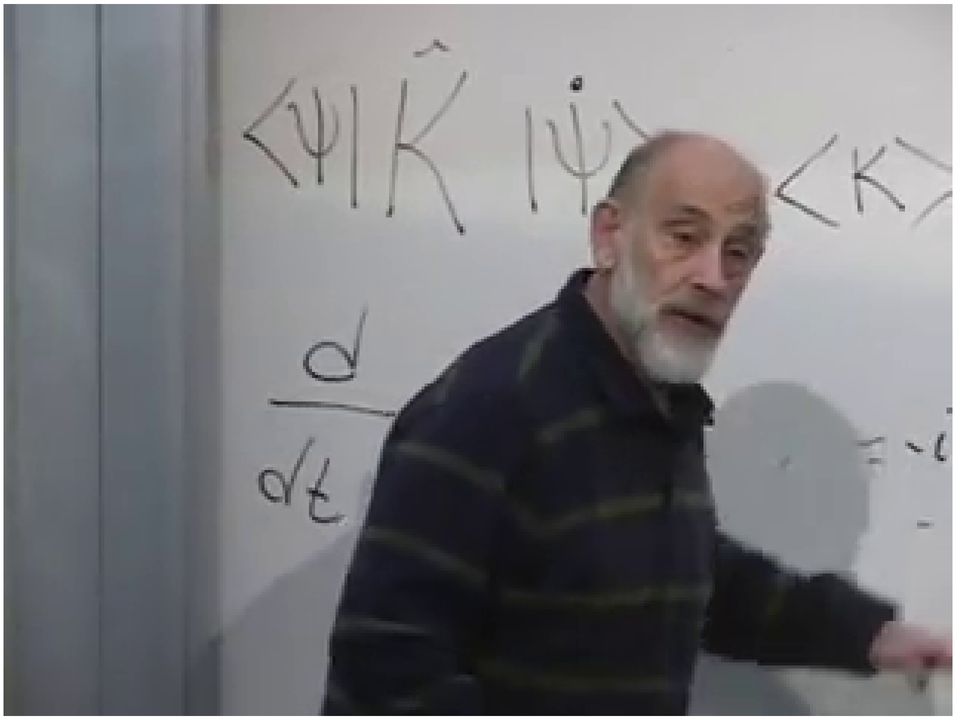

Figure 1.15: The completed commutator form of the expectation-value evolution law.
Lecture frame: 01:16:37.

Question from the audience. Is the result written with $[\hat{K}, \hat{H}]$ or $[\hat{H}, \hat{K}]$, and which sign belongs in front? [a]

Response. Both forms are equivalent because $[\hat{H}, \hat{K}] = -[\hat{K}, \hat{H}]$. With $[\hat{K}, \hat{H}] = \hat{K}\hat{H} - \hat{H}\hat{K}$,

$$\frac{d}{dt}\left\langle \hat{K} \right\rangle = -\frac{i}{\hbar}\left\langle [\hat{K}, \hat{H}] \right\rangle .$$

Writing the commutator in the opposite order changes the prefactor to $+i/\hbar$.

[a]Lecture timestamp: 01:14:46–01:15:18.

This is the quantum counterpart of Equation (1.6). In classical mechanics, take a Poisson bracket with H; in quantum mechanics, take a commutator with $\hat{H}$. The precise correspondence is

$$\{F, H\}_{\mathrm{PB}} \quad \longleftrightarrow \quad \frac{1}{i\hbar}[\hat{F}, \hat{H}] \tag{1.53}$$

in the appropriate classical limit.

1.7 Conserved Quantities

If

$$[\hat{K}, \hat{H}] = 0, \tag{1.54}$$

then Equation (1.52) gives

$$\frac{d}{dt}\left\langle \hat{K} \right\rangle = 0 \tag{1.55}$$

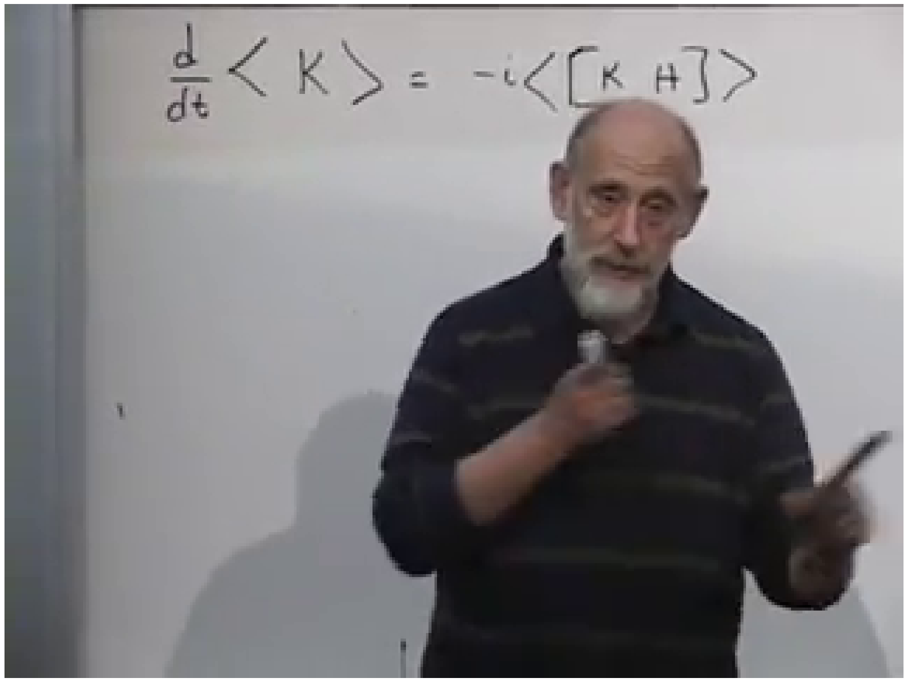

Figure 1.16: A vanishing commutator with the Hamiltonian is the criterion for a time-independent observable distribution.

Lecture frame: 01:18:49.

in every state. More strongly, the spectral projectors of $\hat{K}$ commute with $\hat{H}$, so the full probability distribution of K is time-independent.

The converse requires the same scope: if the expectation value is constant for every state, and $\hat{K}$ has no explicit time dependence, then $[\hat{K}, \hat{H}] = 0$. Constancy in one specially chosen state alone would not be enough.

The Hamiltonian commutes with itself:

$$[\hat{H}, \hat{H}] = 0. \tag{1.56}$$

Therefore a time-independent Hamiltonian is conserved. Since it represents energy, this is energy conservation. The corresponding classical statement is

$$\{H, H\}_{\mathrm{PB}} = 0. \tag{1.57}$$

If the Hamiltonian depends explicitly on time, an additional term $\left\langle \partial \hat{H}/\partial t \right\rangle$ appears and energy need not be conserved. The lecture's argument assumes no such explicit dependence.

1.8 A Free Particle on a Line

1.8.1 From an Abstract Ket to a Wave Function

For a particle on a line, the state may be represented by $\psi(x, t)$. Its position probability density is

$$\rho(x, t) = |\psi(x, t)|^2. \tag{1.58}$$

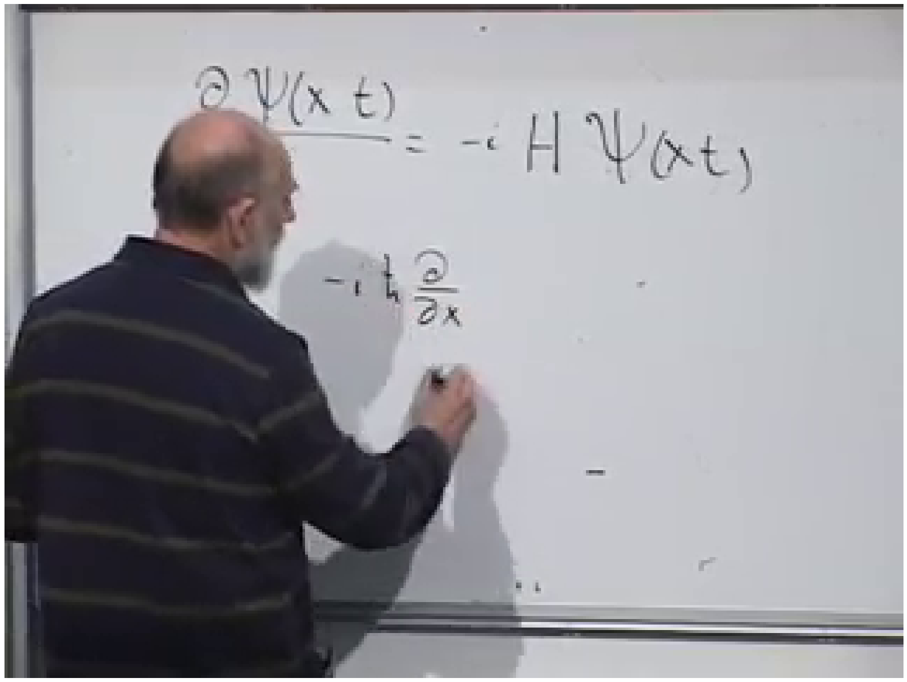

Figure 1.17: The wave-function Schrödinger equation is paired with the differential representation of momentum.

Lecture frame: 01:24:03.

The Fourier transform $\widetilde{\psi}(p,t)$ gives the momentum representation, with $|\widetilde{\psi}(p,t)|^2$ as the momentum probability density. These are not new kinds of state; they are coordinate representations of the same abstract vector.

Multiplication by x and differentiation with respect to x are linear operators on the vector space of wave functions:

$$\hat{x}\psi = x\psi, \qquad \hat{p}\psi = -i\hbar\frac{\partial\psi}{\partial x}. \tag{1.59}$$

For a force-free particle, classical mechanics suggests

$$H = \frac{p^2}{2m}. \tag{1.60}$$

This does not prove the quantum rule, but it gives the correspondence postulate

$$\hat{H} = \frac{\hat{p}^2}{2m}. \tag{1.61}$$

Acting twice with momentum,

$$\hat{p}^2\psi = \left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial\psi}{\partial x}\right) \tag{1.62}$$

$$= -\hbar^2\frac{\partial^2\psi}{\partial x^2}. \tag{1.63}$$

Consequently

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}. \tag{1.64}$$

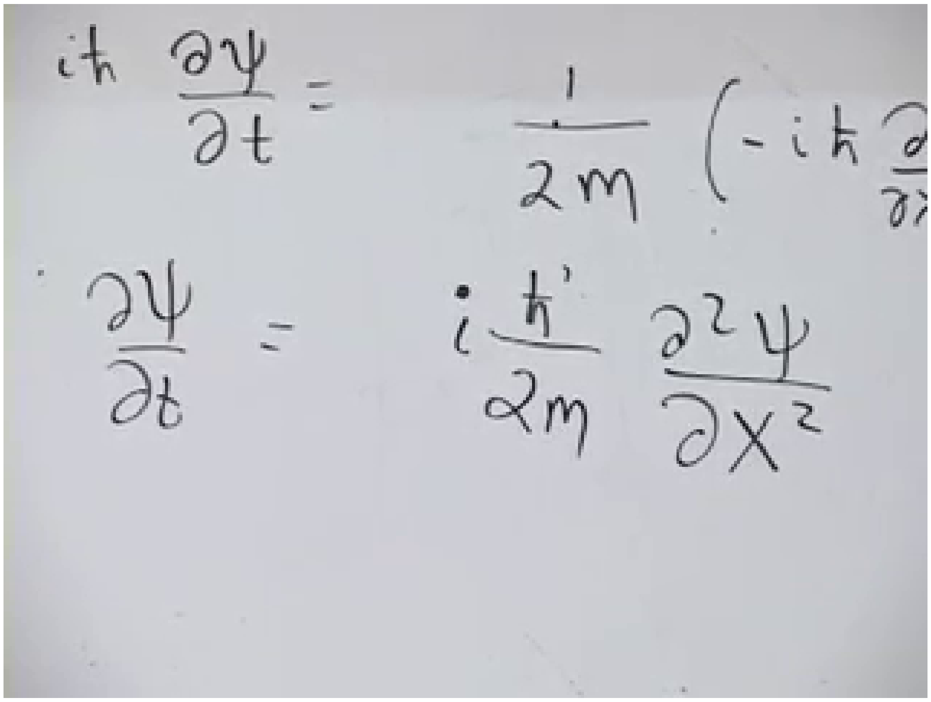

Figure 1.18: The complete free-particle Schrödinger equation with its $\hbar$, i, and minus-sign factors restored.

Lecture frame: 01:30:44.

Question from the audience. Where do the minus sign and the powers of $\hbar$ go in the free-particle equation? [a]

Response. Keep the abstract equation and momentum operator visible at the same time:

$$i\hbar\,\partial_t\psi = \hat{H}\psi, \qquad \hat{H} = \frac{\hat{p}^2}{2m}, \qquad \hat{p} = -i\hbar\,\partial_x.$$

Squaring $\hat{p}$ produces $(-i)^2\hbar^2\partial_x^2 = -\hbar^2\partial_x^2$. No additional factor of i belongs to $\hat{H}$.

[a]Lecture timestamp: 01:27:15–01:29:09.

Substitution into the abstract equation yields Schrödinger's free-particle wave equation:

$$\boxed{i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2}.} \tag{1.65}$$

The Hamiltonian is Hermitian on an appropriate domain because $\hat{p}$ is Hermitian there and the square of a Hermitian operator is Hermitian. Boundary conditions and domains matter in a full treatment; on the line, wave functions are taken to decay sufficiently or are handled with the standard self-adjoint momentum domain.

1.8.2 A Momentum Eigenstate

Look for a solution that remains a momentum eigenstate:

$$\psi_p(x,t) = f(t)e^{ipx/\hbar}. \tag{1.66}$$

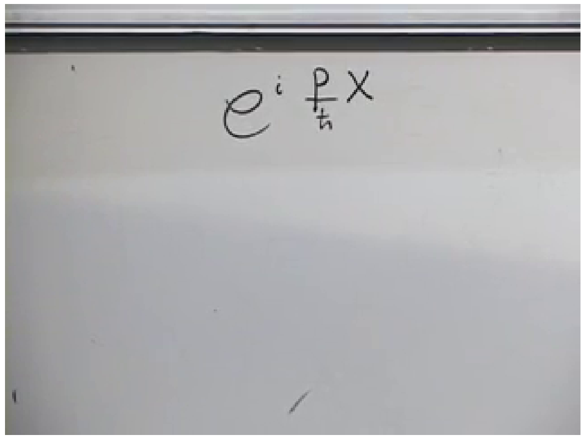

Figure 1.19: The spatial factor $e^{ipx/\hbar}$ is an eigenfunction of the momentum operator.
Lecture frame: 01:32:24.

The number p is the momentum eigenvalue, because

$$-i\hbar\frac{\partial}{\partial x}e^{ipx/\hbar} = p\, e^{ipx/\hbar}. \tag{1.67}$$

The function $f(t)$ is a spectator under spatial differentiation and contains all time dependence. Substituting Equation (1.66) into Equation (1.65) gives

$$i\hbar\dot{f}(t) = \frac{p^2}{2m}f(t). \tag{1.68}$$

Thus

$$f(t) = f(0)\exp\left(-\frac{i}{\hbar}\frac{p^2}{2m}t\right), \tag{1.69}$$

and the full solution is

$$\boxed{\psi_p(x,t) = C\exp\left[\frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)\right].} \tag{1.70}$$

The energy is $E = p^2/(2m)$, so the familiar phase is $(px - Et)/\hbar$. A single ideal momentum eigenstate changes only by an energy phase and retains definite momentum. Such a plane wave is delta-normalized rather than square-integrable; a physical localized state is a superposition of these waves. Its different momentum components acquire different phases, producing motion and spreading, which is the next step in the analysis.

Finally,

$$[\hat{p}, \hat{H}] = \left[\hat{p}, \frac{\hat{p}^2}{2m}\right] = 0. \tag{1.71}$$

Momentum is therefore conserved for a free particle. The classical Hamiltonian flow, unitary state evolution, polarization interference, commutator conservation law, and free-particle wave equation are all different expressions of the same organizing principle: the Hamiltonian generates change with time.

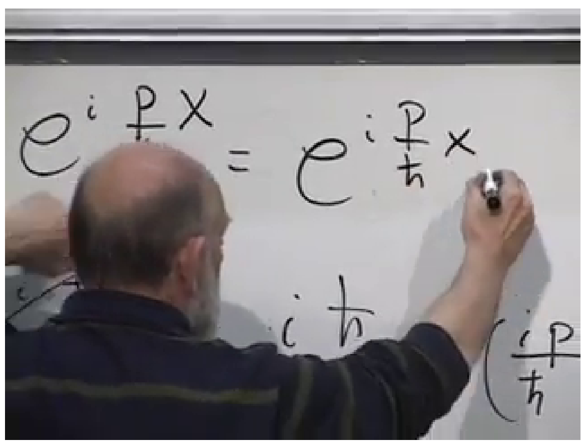

Figure 1.20: The time factor and spatial momentum eigenfunction combine into the free plane wave.

Lecture frame: 01:36:59.